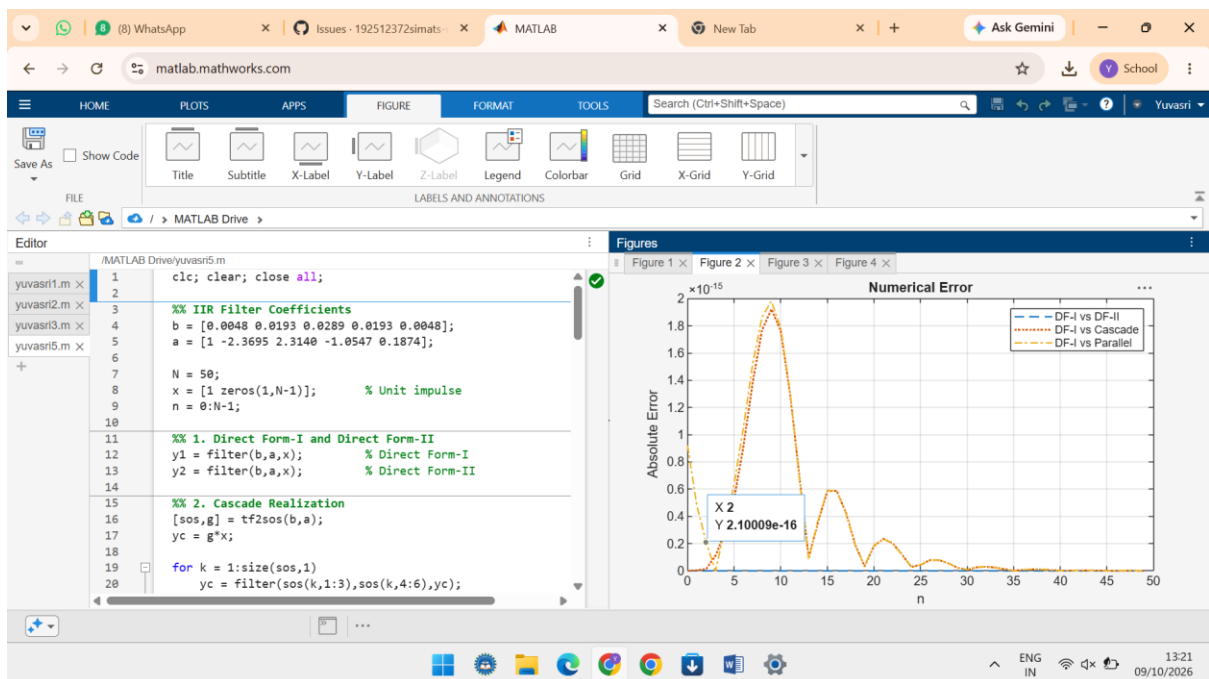
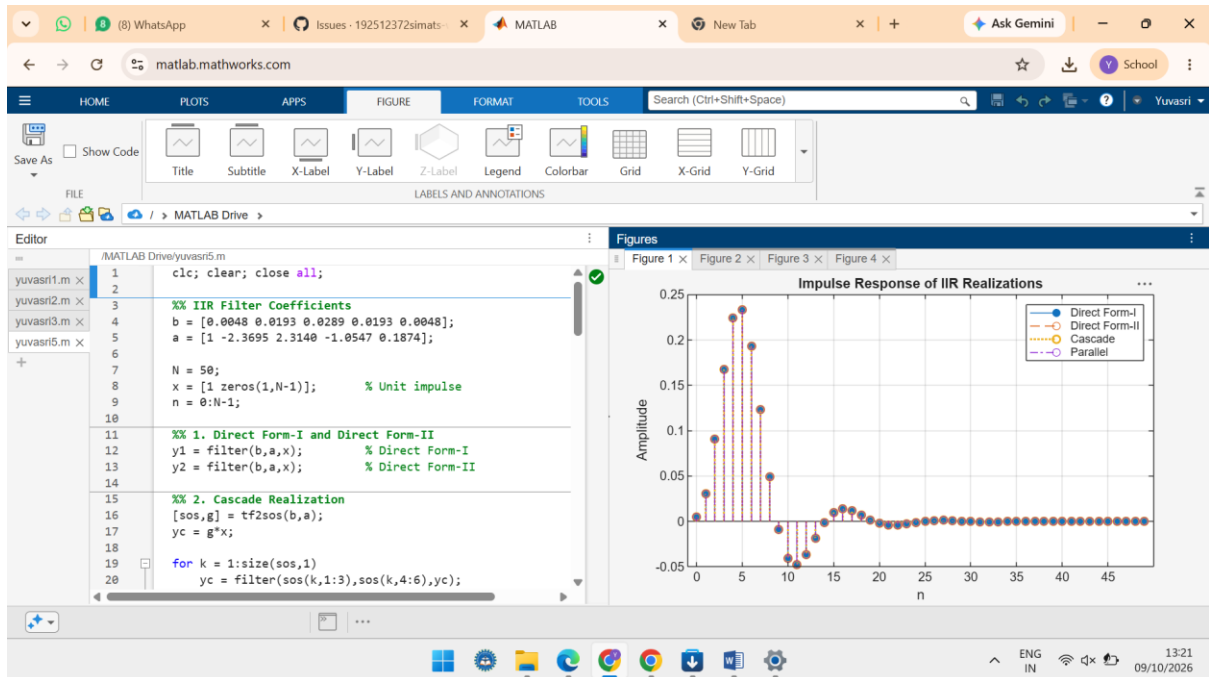
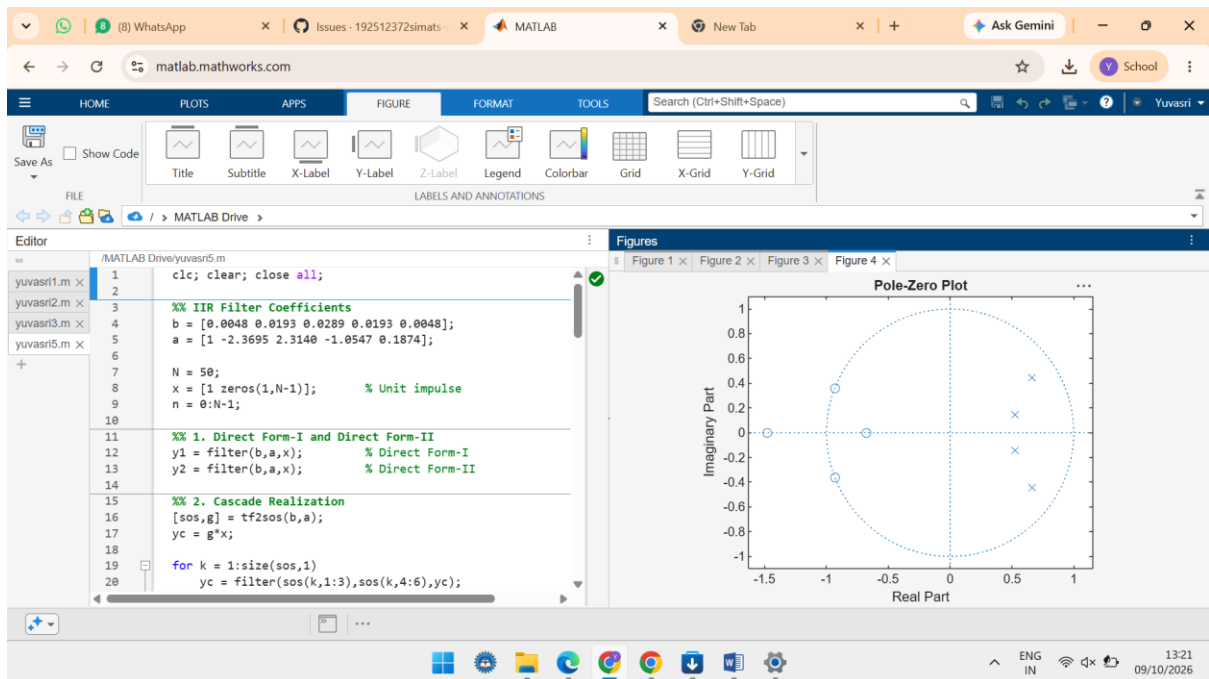
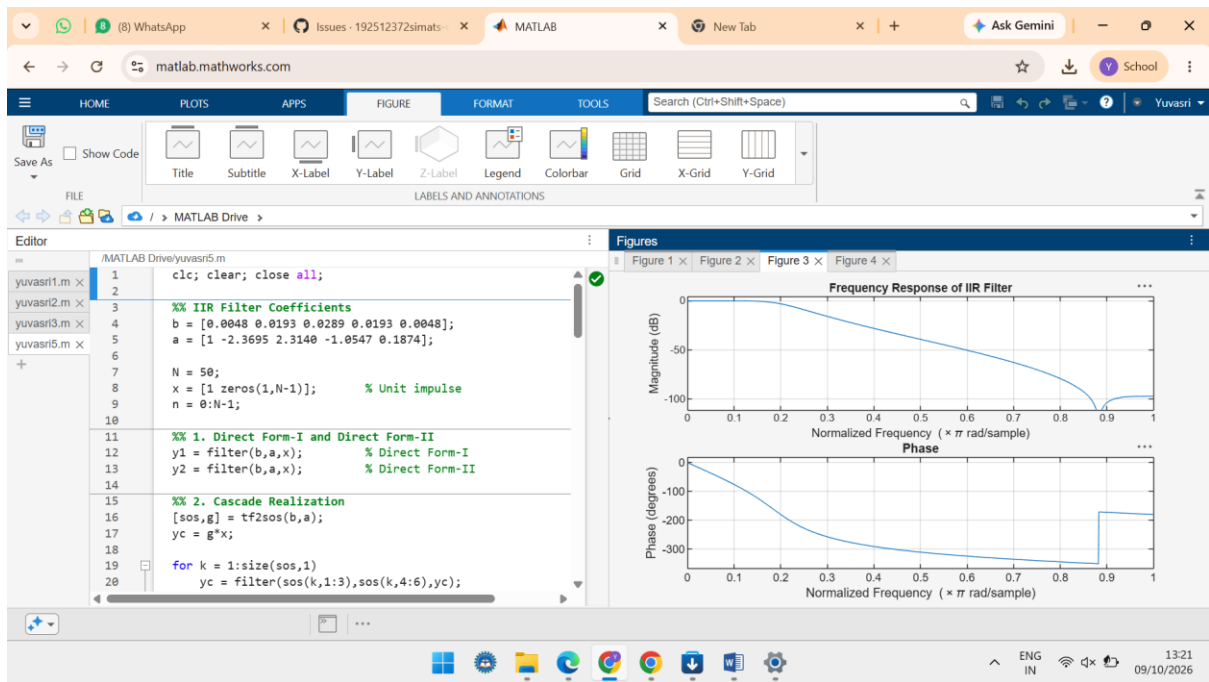


5. Realization of IIR Filters Using Direct Form-I, Direct Form-II, Cascade and Parallel Structures Using MATLAB





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Labels and Annotations: Title Subtitle X-Label Y-Label Z-Label Legend Colorbar Grid X-Grid Y-Grid

yuvasri1.m yuvasri2.m yuvasri3.m yuvasri5.m Figure 1 Figure 2 Figure 3 Figure 4

Command Window

```

Numerator coefficients b =
    0.0048    0.0193    0.0289    0.0193    0.0048

Denominator coefficients a =
    1.0000   -2.3695    2.3140   -1.0547    0.1874

Second Order Sections (Cascade) =
    1.0000    2.1551    1.0000    1.0000   -1.0486    0.2962
    1.0000    1.8657    1.0000    1.0000   -1.3209    0.6327

Cascade Gain =
    0.0048

Residues (Parallel) =
   -0.2698 + 0.1343i
   -0.2698 - 0.1343i
    0.2594 - 0.8024i
    0.2594 + 0.8024i

Poles (Parallel) =

```

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Save As Show Code

FILE

Labels and Annotations: Title Subtitle X-Label Y-Label Z-Label Legend Colorbar Grid X-Grid Y-Grid

yuvasri1.m yuvasri2.m yuvasri3.m yuvasri5.m Figure 1 Figure 2 Figure 3 Figure 4

Command Window

```

    0.2594 - 0.8024i
    0.2594 + 0.8024i

Poles (Parallel) =
    0.6604 + 0.4433i
    0.6604 - 0.4433i
    0.5243 + 0.1459i
    0.5243 - 0.1459i

Direct term =
    0.0256

Comparison Table:


| Structure  | Maximum_Error | Deviation_percent |
|------------|---------------|-------------------|
| "DF-I"     | 0             | 0                 |
| "DF-II"    | 0             | 0                 |
| "Cascade"  | 1.9221e-15    | 8.242e-13         |
| "Parallel" | 1.9811e-15    | 8.4949e-13        |


```

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